

Лабораторная работа

Номер 8

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Информация

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Изучение принципов маршрутизации в IPv4- и IPv6-сетях и принципов настройки сетевого оборудования

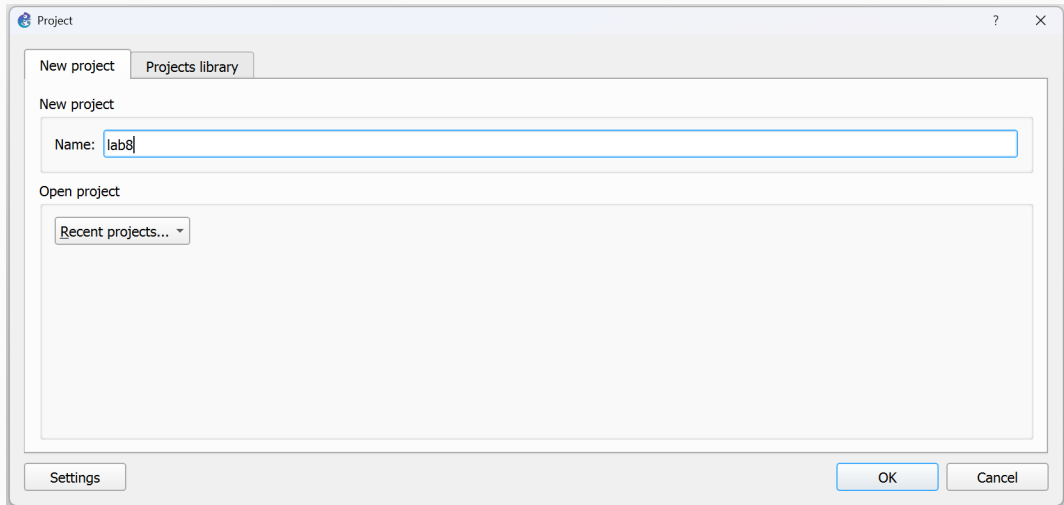


Рис. 1: Создание проекта в GNS3

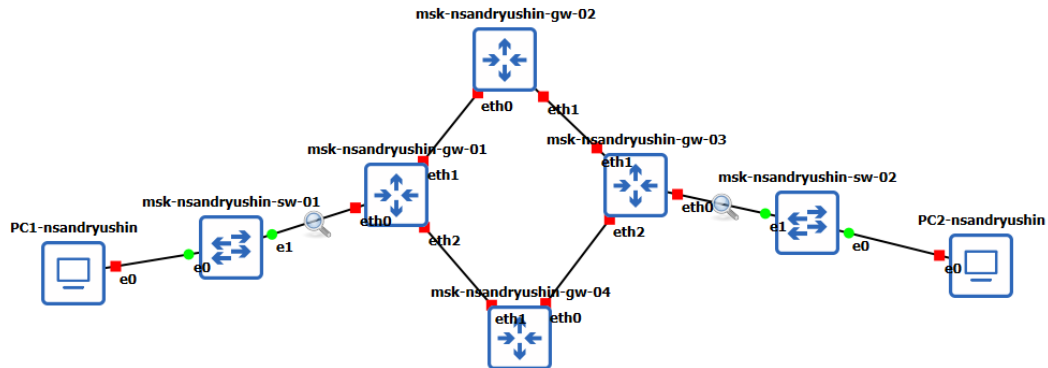
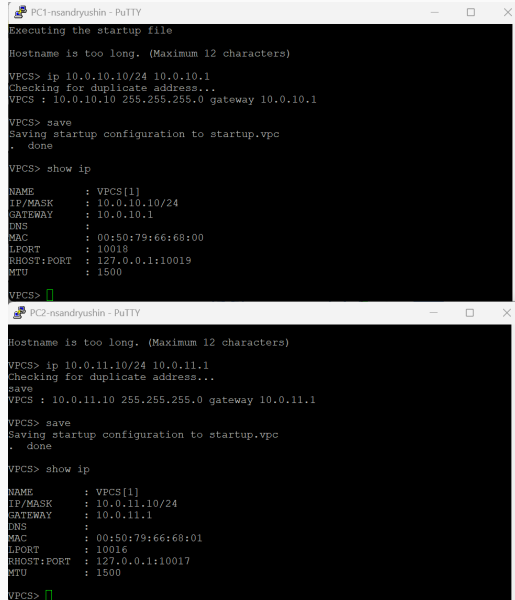


Рис. 2: Топология сети в GNS3



```
PC1-nsandryushin - PuTTY
Executing the startup file

Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.10.10/24 10.0.10.1
Checking for duplicate address...
VPCS : 10.0.10.10 255.255.255.0 gateway 10.0.10.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.10.10/24
GATEWAY    : 10.0.10.1
DNS        :
MAC        : 00:50:79:66:68:00
LPORT      : 10018
RHOST:PORT : 127.0.0.1:10019
MTU        : 1500

VPCS>

PC2-nsandryushin - PuTTY

Hostname is too long. (Maximum 12 characters)

VPCS> ip 10.0.11.10/24 10.0.11.1
Checking for duplicate address...
save
VPCS : 10.0.11.10 255.255.255.0 gateway 10.0.11.1

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ip

NAME       : VPCS[1]
IP/MASK    : 10.0.11.10/24
GATEWAY    : 10.0.11.1
DNS        :
MAC        : 00:50:79:66:68:01
LPORT      : 10016
RHOST:PORT : 127.0.0.1:10017
MTU        : 1500

VPCS>
```

Рис. 3: Настройка IPv4 на PC1

```
frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-01
msk-nsandryushin-gw-01(config)# interface eth0
msk-nsandryushin-gw-01(config-if)# ip address 10.0.10.1/24
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth1
msk-nsandryushin-gw-01(config-if)# ip address 10.0.1.1/24
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth2
msk-nsandryushin-gw-01(config-if)# ip address 10.0.4.2/24
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
exit
!
interface eth1
 ip address 10.0.1.1/24
exit
!
interface eth2
 ip address 10.0.4.2/24
exit
!
end
msk-nsandryushin-gw-01#
```

Рис. 4: Настройка IPv4 на PC2


```
frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-02
msk-nsandryushin-gw-02(config)# interface eth0
msk-nsandryushin-gw-02(config-if)# ip address 10.0.1.2/24
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# interface eth1
msk-nsandryushin-gw-02(config-if)# ip address 10.0.2.1/24
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
exit
!
interface eth1
 ip address 10.0.2.1/24
exit
!
end
msk-nsandryushin-gw-02#
```

Рис. 5: Базовая настройка gw-01

```
frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-03
msk-nsandryushin-gw-03(config)# interface eth0
msk-nsandryushin-gw-03(config-if)# ip address 10.0.11.1/24
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth1
msk-nsandryushin-gw-03(config-if)# ip address 10.0.2.2/24
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth2
msk-nsandryushin-gw-03(config-if)# ip address 10.0.3.1/24
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# show running-config
Building configuration...

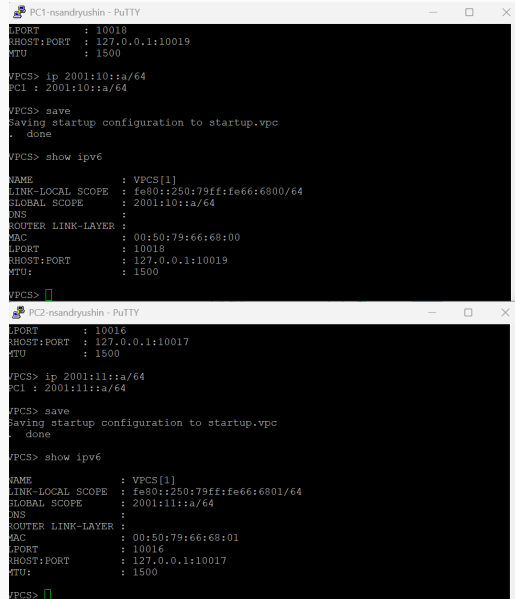
Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
exit
!
interface eth1
 ip address 10.0.2.2/24
exit
!
interface eth2
 ip address 10.0.3.1/24
exit
!
end
msk-nsandryushin-gw-03#
```

Рис. 6: Базовая настройка gw-02

```
frr# configure terminal
frr(config)# hostname msk-nsandryushin-gw-04
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# ip address 10.0.3.2/24
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# interface eth1
msk-nsandryushin-gw-04(config-if)# ip address 10.0.4.1/24
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
exit
!
interface eth1
 ip address 10.0.4.1/24
exit
!
end
msk-nsandryushin-gw-04#
```

Рис. 7: Базовая настройка gw-03



The image displays two screenshots of PuTTY terminal windows, labeled PC1 and PC2, showing the configuration of a VPCS (Virtual Packet Capture System) for a gateway (gw-04).

PC1-nsandryushin - PuTTY

```
LPOR:      : 10018
RHOST:PORT : 127.0.0.1:10019
MTU        : 1500

VPCS> ip 2001:10::a/64
PC1 : 2001:10::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6800/64
GLOBAL SCOPE        : 2001:10::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:00
LPOR                : 10018
RHOST:PORT          : 127.0.0.1:10019
MTU                 : 1500

VPCS>
```

PC2-nsandryushin - PuTTY

```
LPOR:      : 10016
RHOST:PORT : 127.0.0.1:10017
MTU        : 1500

VPCS> ip 2001:11::a/64
PC1 : 2001:11::a/64

VPCS> save
Saving startup configuration to startup.vpc
. done

VPCS> show ipv6

NAME                : VPCS[1]
LINK-LOCAL SCOPE    : fe80::250:79ff:fe66:6801/64
GLOBAL SCOPE        : 2001:11::a/64
DNS                 :
ROUTER LINK-LAYER   :
MAC                 : 00:50:79:66:68:01
LPOR                : 10016
RHOST:PORT          : 127.0.0.1:10017
MTU                 : 1500

VPCS>
```

Рис. 8: Базовая настройка gw-04

```

msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# ipv6 forwarding
msk-nsandryushin-gw-01(config)# interface eth0
msk-nsandryushin-gw-01(config-if)# ipv6 address 2001:10::1/64
msk-nsandryushin-gw-01(config-if)# no ipv6 nd suppress-ra
msk-nsandryushin-gw-01(config-if)# ipv6 nd prefix 2001:10::/64
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth1
msk-nsandryushin-gw-01(config-if)# ipv6 address 2001:1::1/64
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth2
msk-nsandryushin-gw-01(config-if)# ipv6 address 2001:4::2/64
msk-nsandryushin-gw-01(config-if)# no shutdown
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-01
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.10.1/24
 ipv6 address 2001:10::1/64
 ipv6 nd prefix 2001:10::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.1.1/24
 ipv6 address 2001:1::1/64
exit
!
interface eth2
 ip address 10.0.4.2/24
 ipv6 address 2001:4::2/64
exit
!
end
msk-nsandryushin-gw-01#

```

Рис. 9: Настройка IPv6 на PC1

```
msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# ipv6 forwarding
msk-nsandryushin-gw-02(config)# interface eth0
msk-nsandryushin-gw-02(config-if)# ipv6 address 2001:1::2/64
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# interface eth1
msk-nsandryushin-gw-02(config-if)# ipv6 address 2001:2::1/64
msk-nsandryushin-gw-02(config-if)# no shutdown
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-02
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.1.2/24
 ipv6 address 2001:1::2/64
exit
!
interface eth1
 ip address 10.0.2.1/24
 ipv6 address 2001:2::1/64
exit
!
end
msk-nsandryushin-gw-02#
```

Рис. 10: Настройка IPv6 на PC2

```

msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# ipv6 forwarding
msk-nsandryushin-gw-03(config)# interface eth0
msk-nsandryushin-gw-03(config-if)# ipv6 address 2001:11::1/64
msk-nsandryushin-gw-03(config-if)# no ipv6 nd suppress-ra
msk-nsandryushin-gw-03(config-if)# ipv6 nd prefix 2001:11::/64
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth1
msk-nsandryushin-gw-03(config-if)# ipv6 address 2001:2::2/64
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth2
msk-nsandryushin-gw-03(config-if)# ipv6 address 2001:3::1/64
msk-nsandryushin-gw-03(config-if)# no shutdown
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-03
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.11.1/24
 ipv6 address 2001:11::1/64
 ipv6 nd prefix 2001:11::/64
 no ipv6 nd suppress-ra
exit
!
interface eth1
 ip address 10.0.2.2/24
 ipv6 address 2001:2::2/64
exit
!
interface eth2
 ip address 10.0.3.1/24
 ipv6 address 2001:3::1/64
exit
!
end
msk-nsandryushin-gw-03# █

```

Рис. 11: Настройка IPv6 на gw-01

```
msk-nsandryushin-gw-04# configure terminal
msk-nsandryushin-gw-04(config)# ipv6 forwarding
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# ipv6 address 2001:3::2/64
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# interface eth1
msk-nsandryushin-gw-04(config-if)# ipv6 address 2001:4::1/64
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04# show running-config
Building configuration...

Current configuration:
!
frr version 8.2.2
frr defaults traditional
hostname frr
hostname msk-nsandryushin-gw-04
service integrated-vtysh-config
!
interface eth0
 ip address 10.0.3.2/24
 ipv6 address 2001:3::2/64
exit
!
interface eth1
 ip address 10.0.4.1/24
 ipv6 address 2001:4::1/64
exit
!
end
msk-nsandryushin-gw-04#
```

Рис. 12: Настройка IPv6 на gw-02


```
msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router rip
msk-nsandryushin-gw-01(config-router)# version 2
msk-nsandryushin-gw-01(config-router)# network eth0
msk-nsandryushin-gw-01(config-router)# network eth1
msk-nsandryushin-gw-01(config-router)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# █
```

Рис. 13: Настройка IPv6 на gw-03

```
msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# ipv6 forwarding
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# router rip
msk-nsandryushin-gw-02(config-router)# version 2
msk-nsandryushin-gw-02(config-router)# network eth0
msk-nsandryushin-gw-02(config-router)# network eth1
msk-nsandryushin-gw-02(config-router)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# █
```

Рис. 14: Настройка IPv6 на gw-04

```
msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router rip
msk-nsandryushin-gw-03(config-router)# version 2
msk-nsandryushin-gw-03(config-router)# network eth0
msk-nsandryushin-gw-03(config-router)# network eth1
msk-nsandryushin-gw-03(config-router)# network eth2
msk-nsandryushin-gw-03(config-router)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03#
```

Рис. 15: Настройка RIPv2 на gw-01

```
msk-nsandryushin-gw-04# configure terminal
msk-nsandryushin-gw-04(config)# router rip
msk-nsandryushin-gw-04(config-router)# version 2
msk-nsandryushin-gw-04(config-router)# network eth0
msk-nsandryushin-gw-04(config-router)# network eth1
msk-nsandryushin-gw-04(config-router)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04#
```

Рис. 16: Настройка RIPv2 на gw-02

```

msk-nsandryushin-gw-01# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
        O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
        T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
        f - OpenFabric,
        > - selected route, * - FIB route, q - queued, r - rejected, b - backup
        t - trapped, o - offload failure

R>* 10.0.2.0/24 [120/2] via 10.0.1.2, eth1, weight 1, 00:02:30
R>* 10.0.3.0/24 [120/2] via 10.0.4.1, eth2, weight 1, 00:00:10
R>* 10.0.11.0/24 [120/3] via 10.0.1.2, eth1, weight 1, 00:01:28
msk-nsandryushin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24       0.0.0.0           1 self              0
R(n) 10.0.2.0/24       10.0.1.2          2 10.0.1.2           0 02:58
R(n) 10.0.3.0/24       10.0.4.1          2 10.0.4.1           0 02:45
C(i) 10.0.4.0/24       0.0.0.0           1 self              0
C(i) 10.0.10.0/24      0.0.0.0           1 self              0
R(n) 10.0.11.0/24      10.0.1.2          3 10.0.1.2           0 02:58
msk-nsandryushin-gw-01# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 7 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv   Key-chain
    eth0           2      2
    eth1           2      2
    eth2           2      2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway      BadPackets BadRoutes  Distance Last Update
    10.0.1.2      0           0         120     00:00:04
    10.0.4.1      0           0         120     00:00:18
  Distance: (default is 120)
msk-nsandryushin-gw-01#

```

Рис. 17: Настройка RIPv2 на gw-03

```

msk-nsandryushin-gw-02# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.3.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:02:46
R>* 10.0.4.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:03:29
R>* 10.0.10.0/24 [120/2] via 10.0.1.1, eth0, weight 1, 00:03:29
R>* 10.0.11.0/24 [120/2] via 10.0.2.2, eth1, weight 1, 00:02:27
msk-nsandryushin-gw-02# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

        Network          Next Hop          Metric From          Tag Time
C(i) 10.0.1.0/24          0.0.0.0            1 self                0
C(i) 10.0.2.0/24          0.0.0.0            1 self                0
R(n) 10.0.3.0/24          10.0.2.2           2 10.0.2.2             0 02:32
R(n) 10.0.4.0/24          10.0.1.1           2 10.0.1.1             0 02:45
R(n) 10.0.10.0/24         10.0.1.1           2 10.0.1.1             0 02:45
R(n) 10.0.11.0/24         10.0.2.2           2 10.0.2.2             0 02:32
msk-nsandryushin-gw-02# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 1 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv   Key-chain
    eth0           2      2
    eth1           2      2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway      BadPackets BadRoutes Distance Last Update
    10.0.1.1          0          0       120    00:00:18
    10.0.2.2          0          0       120    00:00:01
  Distance: (default is 120)
msk-nsandryushin-gw-02# █

```

Рис. 18: Настройка RIPv2 на gw-04

```

msk-nsandryushin-gw-03# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - PBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.2.1, eth1, weight 1, 00:03:17
R>* 10.0.4.0/24 [120/2] via 10.0.3.2, eth2, weight 1, 00:02:04
R>* 10.0.10.0/24 [120/3] via 10.0.2.1, eth1, weight 1, 00:03:17
msk-nsandryushin-gw-03# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

      Network          Next Hop          Metric From          Tag Time
R(n) 10.0.1.0/24       10.0.2.1           2 10.0.2.1           0 02:57
C(i) 10.0.2.0/24       0.0.0.0            1 self              0
C(i) 10.0.3.0/24       0.0.0.0            1 self              0
R(n) 10.0.4.0/24       10.0.3.2           2 10.0.3.2           0 02:47
R(n) 10.0.10.0/24      10.0.2.1           3 10.0.2.1           0 02:57
C(i) 10.0.11.0/24      0.0.0.0            1 self              0
msk-nsandryushin-gw-03# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 2 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv   Key-chain
    eth0           2      2
    eth1           2      2
    eth2           2      2
  Routing for Networks:
    eth0
    eth1
    eth2
  Routing Information Sources:
    Gateway         BadPackets BadRoutes  Distance Last Update
    10.0.2.1         0           0         120     00:00:05
    10.0.3.2         0           0         120     00:00:15
  Distance: (default is 120)
msk-nsandryushin-gw-03#

```

Рис. 19: Проверка RIP на gw-01

```

msk-nsandryushin-gw-04# show ip route rip
Codes: K - kernel route, C - connected, S - static, R - RIP,
       O - OSPF, I - IS-IS, B - BGP, E - EIGRP, N - NHRP,
       T - Table, v - VNC, V - VNC-Direct, A - Babel, F - FBR,
       f - OpenFabric,
       > - selected route, * - FIB route, q - queued, r - rejected, b - backup
       t - trapped, o - offload failure

R>* 10.0.1.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:32
R>* 10.0.2.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:36
R>* 10.0.10.0/24 [120/2] via 10.0.4.2, eth1, weight 1, 00:02:32
R>* 10.0.11.0/24 [120/2] via 10.0.3.1, eth0, weight 1, 00:02:36
msk-nsandryushin-gw-04# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
        (n) - normal, (s) - static, (d) - default, (r) - redistribute,
        (i) - interface

        Network          Next Hop          Metric From          Tag Time
R(n) 10.0.1.0/24         10.0.4.2          2 10.0.4.2           0 02:52
R(n) 10.0.2.0/24         10.0.3.1          2 10.0.3.1           0 02:58
C(i) 10.0.3.0/24         0.0.0.0           1 self              0
C(i) 10.0.4.0/24         0.0.0.0           1 self              0
R(n) 10.0.10.0/24        10.0.4.2          2 10.0.4.2           0 02:52
R(n) 10.0.11.0/24        10.0.3.1          2 10.0.3.1           0 02:58
msk-nsandryushin-gw-04# show ip rip status
Routing Protocol is "rip"
  Sending updates every 30 seconds with +/-50%, next due in 9 seconds
  Timeout after 180 seconds, garbage collect after 120 seconds
  Outgoing update filter list for all interface is not set
  Incoming update filter list for all interface is not set
  Default redistribution metric is 1
  Redistributing:
  Default version control: send version 2, receive version 2
    Interface      Send Recv   Key-chain
    eth0           2      2
    eth1           2      2
  Routing for Networks:
    eth0
    eth1
  Routing Information Sources:
    Gateway         BadPackets BadRoutes  Distance Last Update
    10.0.3.1         0           0         120      00:00:03
    10.0.4.2         0           0         120      00:00:10
  Distance: (default is 120)
msk-nsandryushin-gw-04#

```

Рис. 20: Проверка RIP на gw-02


```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=15.337 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=6.752 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=18.950 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=7.272 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=4.126 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    4.297 ms  1.364 ms  0.746 ms
 2  10.0.4.1     5.317 ms  4.480 ms  2.347 ms
 3  10.0.3.1     7.541 ms  7.441 ms  3.720 ms
 4  10.0.11.10   7.358 ms  4.779 ms  2.758 ms

VPCS> █
```

Рис. 21: Проверка RIP на gw-03

```

msk-nsandryushin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface

```

	Network	Next Hop	Metric	From	Tag	Time
C(i)	10.0.1.0/24	0.0.0.0	1	self	0	
R(n)	10.0.2.0/24	10.0.1.2	2	10.0.1.2	0	02:42
R(n)	10.0.3.0/24	10.0.4.1	2	10.0.4.1	0	02:34
C(i)	10.0.4.0/24	0.0.0.0	1	self	0	
C(i)	10.0.10.0/24	0.0.0.0	1	self	0	
R(n)	10.0.11.0/24	10.0.4.1	3	10.0.4.1	0	02:34

```

msk-nsandryushin-gw-01# █

```

Рис. 22: Проверка RIP на gw-04

```
msk-nsandryushin-gw-04# configure terminal
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# shutdown
msk-nsandryushin-gw-04(config-if)#
```

Рис. 23: Проверка связности IPv4

```
msk-nsandryushin-gw-01# show ip rip
Codes: R - RIP, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface
```

	Network	Next Hop	Metric	From	Tag	Time
C(i)	10.0.1.0/24	0.0.0.0	1	self	0	
R(n)	10.0.2.0/24	10.0.1.2	2	10.0.1.2	0	02:44
R(n)	10.0.3.0/24	10.0.4.1	16	10.0.4.1	0	01:52
C(i)	10.0.4.0/24	0.0.0.0	1	self	0	
C(i)	10.0.10.0/24	0.0.0.0	1	self	0	
R(n)	10.0.11.0/24	10.0.4.1	16	10.0.4.1	0	01:52

```
msk-nsandryushin-gw-01#
```

Рис. 24: Таблица RIP на gw-01

```
msk-nsandryushin-gw-04(config-if)# interface eth0
msk-nsandryushin-gw-04(config-if)# no shutdown
msk-nsandryushin-gw-04(config-if)#
```

Рис. 25: Отключение интерфейса на gw-04

```
VPCS> ping 10.0.11.10
```

```
10.0.11.10 icmp_seq=1 timeout
```

```
10.0.11.10 icmp_seq=2 timeout
```

```
10.0.11.10 icmp_seq=3 timeout
```

```
10.0.11.10 icmp_seq=4 timeout
```

```
10.0.11.10 icmp_seq=5 timeout
```

```
VPCS> trace 10.0.11.10 -P 6
```

```
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
```

```
1  10.0.10.1    2.129 ms  1.936 ms  1.251 ms
```

```
2  10.0.1.2     2.893 ms  2.973 ms  3.188 ms
```

```
3      *      *
```

```
4      *      *
```

```
5      *      *
```

```
6      *      *
```

```
7      *      *
```

```
8      *      *
```

```
VPCS> █
```

Рис. 26: Изменение метрик RIP после сбоя

1 0.000000	fe80::ed5:34ff:fef4...	ff02::1	ICMPv6	110 Router Advertisement from 0c:d5:34:f4:00:00
2 0.854921	10.0.10.1	224.0.0.9	RIPv2	66 Response
3 2.859154	10.0.10.1	224.0.0.9	RIPv2	66 Response
4 28.145901	10.0.10.1	224.0.0.9	RIPv2	146 Response

Рис. 27: Включение интерфейса на gw-04

```
msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router ripng
msk-nsandryushin-gw-01(config-router)# network eth0
msk-nsandryushin-gw-01(config-router)# network eth1
msk-nsandryushin-gw-01(config-router)# network eth2
msk-nsandryushin-gw-01(config-router)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# █
```

Рис. 28: Проверка доступности во время сходимости


```
msk-nsandryushin-gw-02(config-if)# router ripng
msk-nsandryushin-gw-02(config-router)# network eth0
msk-nsandryushin-gw-02(config-router)# network eth1
msk-nsandryushin-gw-02(config-router)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02#
```

Рис. 29: Анализ пакетов RIPv2 в Wireshark

```
msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router ripng
msk-nsandryushin-gw-03(config-router)# network eth0
msk-nsandryushin-gw-03(config-router)# network eth1
msk-nsandryushin-gw-03(config-router)# network eth2
msk-nsandryushin-gw-03(config-router)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03# █
```

Рис. 30: Настройка RIPng на gw-01

```
msk-nsandryushin-gw-04(config-if)# router ripng
msk-nsandryushin-gw-04(config-router)# network eth0
msk-nsandryushin-gw-04(config-router)# network eth1
msk-nsandryushin-gw-04(config-router)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04#
```

Рис. 31: Настройка RIPvng на gw-02

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=11.499 ms
2001:11::a icmp6_seq=2 ttl=58 time=3.296 ms
2001:11::a icmp6_seq=3 ttl=58 time=3.708 ms
2001:11::a icmp6_seq=4 ttl=58 time=3.343 ms
2001:11::a icmp6_seq=5 ttl=58 time=10.094 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    3.054 ms   1.571 ms   1.275 ms
 2 2001:1::2     4.812 ms   3.328 ms   2.749 ms
 3 2001:2::2     9.434 ms   4.926 ms   8.099 ms
 4 2001:11::a    4.816 ms   9.333 ms   6.534 ms

VPCS> 
```

Рис. 32: Настройка RIPng на gw-03

```

msk-nsandryushin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64      ::      self      1      0
R(n) 2001:2::/64      fe80::ed8:daff:fe5f:0  eth1      2      0  02:40
R(n) 2001:3::/64      fe80::e05:bfff:feca:1  eth2      2      0  02:41
C(i) 2001:4::/64      ::      self      1      0
C(i) 2001:10::/64     ::      self      1      0
R(n) 2001:11::/64     fe80::ed8:daff:fe5f:0  eth1      3      0  02:41
msk-nsandryushin-gw-01#

```

Рис. 33: Настройка RIPng на gw-04

```
msk-nsandryushin-gw-04(config)# interface eth0  
msk-nsandryushin-gw-04(config-if)# shutdown  
msk-nsandryushin-gw-04(config-if)#
```

Рис. 34: Проверка связности IPv6

```

msk-nsandryushin-gw-01# show ipv6 ripng
Codes: R - RIPng, C - connected, S - Static, O - OSPF, B - BGP
Sub-codes:
      (n) - normal, (s) - static, (d) - default, (r) - redistribute,
      (i) - interface, (a/S) - aggregated/Suppressed

  Network      Next Hop      Via      Metric Tag Time
C(i) 2001:1::/64      ::      self      1      0
R(n) 2001:2::/64      fe80::ed8:daff:fe5f:0      eth1      2      0      02:38
R(n) 2001:3::/64      fe80::ed8:daff:fe5f:0      eth1      3      0      02:38
C(i) 2001:4::/64      ::      self      1      0
C(i) 2001:10::/64      ::      self      1      0
R(n) 2001:11::/64      fe80::ed8:daff:fe5f:0      eth1      3      0      02:38
msk-nsandryushin-gw-01# █

```

Рис. 35: Таблица RIPng на gw-01

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=6.787 ms
2001:11::a icmp6_seq=2 ttl=58 time=4.883 ms
2001:11::a icmp6_seq=3 ttl=58 time=4.505 ms
2001:11::a icmp6_seq=4 ttl=58 time=5.099 ms
2001:11::a icmp6_seq=5 ttl=58 time=40.125 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    2.622 ms  1.052 ms  0.476 ms
 2 2001:1::2     2.563 ms  1.942 ms  1.982 ms
 3 2001:2::2     5.026 ms  4.869 ms  3.958 ms
 4 2001:11::a    5.500 ms  5.339 ms  6.440 ms

VPCS> █
```

Рис. 36: Отключение интерфейса IPv6 на gw-04


```
msk-nsandryushin-gw-04(config-if)# interface eth0  
msk-nsandryushin-gw-04(config-if)# no shutdown  
msk-nsandryushin-gw-04(config-if)#
```

Рис. 37: Изменения в таблице RIPvng

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=7.159 ms
2001:11::a icmp6_seq=2 ttl=58 time=3.384 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.507 ms
2001:11::a icmp6_seq=4 ttl=58 time=4.649 ms
2001:11::a icmp6_seq=5 ttl=58 time=6.221 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    1.753 ms   0.659 ms   0.531 ms
 2 2001:1::2     13.139 ms  3.201 ms   3.363 ms
 3 2001:2::2     4.364 ms   5.666 ms   4.575 ms
 4 2001:11::a    7.346 ms   5.873 ms   7.205 ms

VPCS> 
```

Рис. 38: Проверка трассировки IPv6 после сбоя

1	0.000000	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=1, hop limit=64 (reply in 2)
2	0.003380	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=1, hop limit=58 (request in 1)
3	1.007774	2001:11::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=2, hop limit=64 (reply in 4)
4	1.010543	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=2, hop limit=58 (request in 3)
5	2.012440	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=3, hop limit=64 (reply in 6)
6	2.016682	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=3, hop limit=58 (request in 5)
7	3.020346	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=4, hop limit=64 (reply in 8)
8	3.024414	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=4, hop limit=58 (request in 7)
9	4.026622	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=5, hop limit=64 (reply in 10)
10	4.030371	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=5, hop limit=58 (request in 9)
11	5.074644	fe80::ed5:34ff:fef4...	2001:10::a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00
12	5.784419	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
13	5.785799	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
14	5.787845	2001:11::a	2001:11::a	UDP	126 37568 → 37569 Len=64
15	5.788581	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
16	5.789058	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
17	5.789470	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
18	5.790696	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
19	5.794221	2001:11::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
20	5.796092	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
21	5.798194	2001:11::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
22	5.800355	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
23	5.803391	2001:11::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
24	5.804815	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
25	5.809429	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
26	5.810863	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
27	5.816224	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
28	5.817297	2001:11::a	2001:11::a	UDP	126 37568 → 37569 Len=64
29	5.820823	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
30	5.821051	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
31	5.828793	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
32	5.830154	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
33	5.835110	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
34	5.836864	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
35	5.843583	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
36	6.113915	fe80::ed5:34ff:fef4...	2001:10::a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00
37	7.154049	fe80::ed5:34ff:fef4...	2001:10::a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00
38	10.600228	10.0.10.1	224.0.0.9	RIPv2	146 Response
39	23.442667	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1

1	0.000000	10.0.11.1	224.0.0.9	RIPv2	146 Response
2	6.135421	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=1, hop limit=61 (reply in 3)
3	6.135655	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=1, hop limit=61 (request in 2)
4	7.143164	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=2, hop limit=61 (reply in 5)
5	7.143398	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=2, hop limit=61 (request in 4)
6	8.148073	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=3, hop limit=61 (reply in 7)
7	8.148267	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=3, hop limit=61 (request in 6)
8	9.155842	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=4, hop limit=61 (reply in 9)
9	9.156143	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=4, hop limit=61 (request in 8)
10	10.162039	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=5, hop limit=61 (reply in 11)
11	10.162368	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=5, hop limit=61 (request in 10)
12	11.153272	fe80::eb5:37ff:fe3f...	2001:11::a	ICMPv6	86 Neighbor Solicitation for 2001:11::a from 0c:b5:37:3f:00:00
13	11.958998	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
14	11.959245	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
15	11.965815	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
16	11.966056	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
17	11.973470	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
18	11.973793	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
19	12.193011	fe80::eb5:37ff:fe3f...	2001:11::a	ICMPv6	86 Neighbor Solicitation for 2001:11::a from 0c:b5:37:3f:00:00
20	13.232563	fe80::eb5:37ff:fe3f...	2001:11::a	ICMPv6	86 Neighbor Solicitation for 2001:11::a from 0c:b5:37:3f:00:00
21	18.072984	fe80::eb5:37ff:fe3f...	ff02::9	RIPng	186 Command Response, Version 1
22	36.006672	10.0.11.1	224.0.0.9	RIPv2	146 Response
23	48.082730	fe80::eb5:37ff:fe3f...	ff02::9	RIPng	186 Command Response, Version 1

Рис. 40: Анализ IPv6 трафика в Wireshark

```
msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router ospf
msk-nsandryushin-gw-01(config-router)# network 10.0.10.0/24 area 0.0.0.0
msk-nsandryushin-gw-01(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-nsandryushin-gw-01(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-nsandryushin-gw-01(config-router)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01#
```

Рис. 41: Настройка OSPF на gw-01

```
msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# router ospf
msk-nsandryushin-gw-02(config-router)# network 10.0.1.0/24 area 0.0.0.0
msk-nsandryushin-gw-02(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-nsandryushin-gw-02(config-router)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02# █
```

Рис. 42: Настройка OSPF на gw-02

```
msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router ospf
msk-nsandryushin-gw-03(config-router)# network 10.0.11.0/24 area 0.0.0.0
msk-nsandryushin-gw-03(config-router)# network 10.0.2.0/24 area 0.0.0.0
msk-nsandryushin-gw-03(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-nsandryushin-gw-03(config-router)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03#
```

Рис. 43: Настройка OSPF на gw-03

```
msk-nsandryushin-gw-04(config-if)# router ospf
msk-nsandryushin-gw-04(config-router)# network 10.0.3.0/24 area 0.0.0.0
msk-nsandryushin-gw-04(config-router)# network 10.0.4.0/24 area 0.0.0.0
msk-nsandryushin-gw-04(config-router)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04#
```

Рис. 44: Настройка OSPF на gw-04


```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=13.670 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=2.687 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=8.155 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=7.790 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=6.081 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    5.541 ms  2.604 ms  4.040 ms
 2  10.0.4.1     9.293 ms  3.198 ms  6.435 ms
 3  10.0.2.2    12.825 ms  7.816 ms  6.390 ms
 4  10.0.11.10  14.574 ms  9.895 ms 13.379 ms

VPCS> 
```

Рис. 45: Проверка связности через OSPF

```

msk-nsandryushin-gw-01# show ip ospf neighbor
Neighbor ID      Pri State                Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1         1 Full/Backup         3m27s            31.783s 10.0.1.2      et
h1:10.0.1.1      0      0      0
10.0.4.1         1 Full/Backup         1m21s            29.990s 10.0.4.1      et
h2:10.0.4.2      0      0      0

msk-nsandryushin-gw-01# show ip ospf route
===== OSPF network routing table =====
N    10.0.1.0/24      [100] area: 0.0.0.0
      directly attached to eth1
N    10.0.2.0/24      [200] area: 0.0.0.0
      via 10.0.1.2, eth1
N    10.0.3.0/24      [200] area: 0.0.0.0
      via 10.0.4.1, eth2
N    10.0.4.0/24      [100] area: 0.0.0.0
      directly attached to eth2
N    10.0.10.0/24     [100] area: 0.0.0.0
      directly attached to eth0
N    10.0.11.0/24     [300] area: 0.0.0.0
      via 10.0.1.2, eth1
      via 10.0.4.1, eth2

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-nsandryushin-gw-01# █

```

Рис. 46: Таблица соседей и маршрутов OSPF на gw-01

```

msk-nsandryushin-gw-01# show ip ospf neighbor

Neighbor ID      Pri State                Up Time           Dead Time Address      In
terface          RXmtL RqstL DBsmL
10.0.2.1         1 Full/Backup          4m01s             37.654s 10.0.1.2          et
h1:10.0.1.1      0      0      0
10.0.4.1         1 Full/Backup          1m55s             35.861s 10.0.4.1          et
h2:10.0.4.2      0      0      0

msk-nsandryushin-gw-01# show ip ospf route
===== OSPF network routing table =====
N    10.0.1.0/24          [100] area: 0.0.0.0
      directly attached to eth1
N    10.0.2.0/24          [200] area: 0.0.0.0
      via 10.0.1.2, eth1
N    10.0.3.0/24          [300] area: 0.0.0.0
      via 10.0.1.2, eth1
N    10.0.4.0/24          [100] area: 0.0.0.0
      directly attached to eth2
N    10.0.10.0/24         [100] area: 0.0.0.0
      directly attached to eth0
N    10.0.11.0/24         [300] area: 0.0.0.0
      via 10.0.1.2, eth1

===== OSPF router routing table =====

===== OSPF external routing table =====

msk-nsandryushin-gw-01# █

```

Рис. 47: Мониторинг OSPF на gw-01

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=4.639 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=6.683 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=4.127 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=3.546 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=6.524 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    1.486 ms  0.439 ms  0.806 ms
 2  10.0.1.2     6.410 ms  1.842 ms  2.354 ms
 3  10.0.2.2     4.067 ms  2.309 ms  4.865 ms
 4  10.0.11.10   4.043 ms  5.901 ms  6.870 ms

VPCS> █
```

Рис. 48: Повторная трассировка IPv4

```
msk-nsandryushin-gw-04 (config-if) # interface eth0  
msk-nsandryushin-gw-04 (config-if) # no shutdown  
msk-nsandryushin-gw-04 (config-if) #
```

Рис. 49: Включение интерфейса на gw-04

```
VPCS> ping 10.0.11.10

84 bytes from 10.0.11.10 icmp_seq=1 ttl=61 time=3.312 ms
84 bytes from 10.0.11.10 icmp_seq=2 ttl=61 time=5.454 ms
84 bytes from 10.0.11.10 icmp_seq=3 ttl=61 time=5.039 ms
84 bytes from 10.0.11.10 icmp_seq=4 ttl=61 time=4.097 ms
84 bytes from 10.0.11.10 icmp_seq=5 ttl=61 time=5.922 ms

VPCS> trace 10.0.11.10 -P 6
trace to 10.0.11.10, 8 hops max (TCP), press Ctrl+C to stop
 1  10.0.10.1    0.851 ms   1.516 ms   0.552 ms
 2  10.0.4.1     3.025 ms   1.893 ms   1.465 ms
 3  10.0.2.2     2.638 ms   5.644 ms   4.798 ms
 4  10.0.11.10   4.833 ms   6.347 ms   4.860 ms

VPCS> 
```

Рис. 50: Трассировка через восстановленный маршрут

1	0.000000	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=1, hop limit=64 (reply in 2)
2	0.003380	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=1, hop limit=58 (request in 1)
3	1.007774	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=2, hop limit=64 (reply in 4)
4	1.010543	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=2, hop limit=58 (request in 3)
5	2.012440	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=3, hop limit=64 (reply in 6)
6	2.016682	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=3, hop limit=58 (request in 5)
7	3.020346	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=4, hop limit=64 (reply in 8)
8	3.024414	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=4, hop limit=58 (request in 7)
9	4.026622	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x2fdb, seq=5, hop limit=64 (reply in 10)
10	4.030371	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x2fdb, seq=5, hop limit=58 (request in 9)
11	5.074644	fe80::ed5:34ff:fef4...	2001:10::a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00
12	5.784419	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
13	5.785799	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
14	5.787845	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
15	5.788581	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
16	5.789058	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
17	5.789470	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
18	5.790696	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
19	5.794221	2001:1::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
20	5.796092	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
21	5.798194	2001:1::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
22	5.800355	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
23	5.803391	2001:1::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
24	5.804815	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
25	5.809429	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
26	5.810863	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
27	5.816224	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
28	5.817297	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
29	5.820823	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
30	5.821051	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
31	5.828793	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
32	5.830154	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
33	5.835110	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
34	5.836864	2001:10::a	2001:11::a	UDP	126 37568 → 37569 Len=64
35	5.843583	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
36	6.113915	fe80::ed5:34ff:fef4...	2001:10::a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00
37	7.154049	fe80::ed5:34ff:fef4...	2001:10::a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00

Рис. 51: Анализ ошибок ICMPv6 в Wireshark

52	195.884674	fe80::ed5:34ff:fef4...	ff02::1	ICMPv6	110 Router Advertisement from 0c:d5:34:f4:00:00
53	199.661651	10.0.10.1	224.0.0.9	RIPv2	146 Response
54	226.920117	10.0.10.1	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.5 for any sources
55	226.921262	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
56	227.192342	10.0.10.1	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.5 for any sources
57	229.519281	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1
58	234.664902	10.0.10.1	224.0.0.9	RIPv2	146 Response
59	236.920946	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
60	246.922178	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
61	256.924322	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
62	263.672737	10.0.10.1	224.0.0.9	RIPv2	146 Response
63	266.925478	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
64	266.933389	10.0.10.1	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.6 for any sources
65	267.994423	10.0.10.1	224.0.0.22	IGMPv3	60 Membership Report / Join group 224.0.0.6 for any sources
66	272.528165	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1
67	276.927290	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
68	286.928528	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
69	294.684427	10.0.10.1	224.0.0.9	RIPv2	146 Response
70	296.931992	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
71	306.935385	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
72	315.540811	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1
73	316.939135	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
74	325.692372	10.0.10.1	224.0.0.9	RIPv2	146 Response
75	326.940765	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
76	336.943400	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
77	346.947587	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
78	348.697735	10.0.10.1	224.0.0.9	RIPv2	146 Response
79	353.549690	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1
80	356.950800	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
81	366.952326	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
82	376.707171	10.0.10.1	224.0.0.9	RIPv2	146 Response
83	376.954518	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
84	386.956580	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
85	388.559679	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1
86	396.960370	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
87	406.719743	10.0.10.1	224.0.0.9	RIPv2	146 Response
88	406.962616	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
89	416.966892	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
90	426.969836	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
91	431.575248	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1
92	436.972058	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
93	437.729903	10.0.10.1	224.0.0.9	RIPv2	146 Response
94	446.974797	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
95	456.977776	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
96	463.255135	Private_66:68:00	Broadcast	ARP	64 Who has 10.0.10.1? Tell 10.0.10.10
97	463.256583	0c:d5:34:f4:00:00	Private_66:68:00	ARP	60 10.0.10.1 is at 0c:d5:34:f4:00:00
→	98 463.258102	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request id=0xfedc, seq=1/256, ttl=64 (reply in 99)

Рис. 52: Анализ OSPF и RIP трафика

97	463.256583	0c:d5:34:f4:00:00	Private_66:68:00	ARP	60	10.0.10.1 is at 0c:d5:34:f4:00:00
98	463.258102	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xfedc, seq=1/256, ttl=64 (reply in 99)
99	463.271539	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xfedc, seq=1/256, ttl=61 (request in 98)
100	464.273975	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0xffdc, seq=2/512, ttl=64 (reply in 101)
101	464.276479	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0xffdc, seq=2/512, ttl=61 (request in 100)
102	465.278281	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x00dd, seq=3/768, ttl=64 (reply in 103)
103	465.285973	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x00dd, seq=3/768, ttl=61 (request in 102)
104	466.287283	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x01dd, seq=4/1024, ttl=64 (reply in 105)
105	466.294795	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x01dd, seq=4/1024, ttl=61 (request in 104)
106	466.738036	10.0.10.1	224.0.0.9	RIPv2	146	Response
107	466.981035	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
108	467.297318	10.0.10.10	10.0.11.10	ICMP	98	Echo (ping) request id=0x02dd, seq=5/1280, ttl=64 (reply in 109)
109	467.302937	10.0.11.10	10.0.10.10	ICMP	98	Echo (ping) reply id=0x02dd, seq=5/1280, ttl=61 (request in 108)
110	468.349414	0c:d5:34:f4:00:00	Private_66:68:00	ARP	60	Who has 10.0.10.10? Tell 10.0.10.1
111	468.349737	Private_66:68:00	0c:d5:34:f4:00:00	ARP	60	10.0.10.10 is at 00:50:79:66:68:00
112	469.879162	10.0.10.10	10.0.11.10	TCP	74	6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251781 TSecr=0 WS...
113	469.884142	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
114	469.884691	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
115	469.886865	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
116	469.888962	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
117	469.892378	10.0.10.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
118	469.894329	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
119	469.903212	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
120	469.903740	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
121	469.906473	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
122	469.909118	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
123	469.914524	10.0.4.1	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
124	469.917170	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
125	469.929172	10.0.2.2	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
126	469.930011	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
127	469.937228	10.0.2.2	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
128	469.937683	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
129	469.943748	10.0.2.2	10.0.10.10	ICMP	102	Time-to-live exceeded (Time to live exceeded in transit)
130	469.945709	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
131	469.959532	10.0.11.10	10.0.10.10	TCP	60	6386 → 6385 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
132	469.960436	10.0.10.10	10.0.11.10	TCP	74	[TCP Retransmission] 6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=...
133	469.970951	10.0.11.10	10.0.10.10	TCP	60	[TCP Port numbers reused] 6386 → 6385 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
134	469.971020	10.0.10.10	10.0.11.10	TCP	74	6385 → 6386 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251781 TSecr=0 WS...
135	469.984117	10.0.11.10	10.0.10.10	TCP	60	[TCP Port numbers reused] 6386 → 6385 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
136	475.590429	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186	Command Response, Version 1
137	476.985317	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet
138	486.988012	10.0.10.1	224.0.0.5	OSPF	78	Hello Packet

Рис. 53: Анализ ICMP и ARP

159	580.267352	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply	id=0x73dd, seq=1/256, ttl=61 (request in 158)
160	581.271187	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) request	id=0x74dd, seq=2/512, ttl=64 (reply in 161)
161	581.277495	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply	id=0x74dd, seq=2/512, ttl=61 (request in 160)
162	581.783442	10.0.10.1	224.0.0.9	RIPv2	146 Response	
163	582.280220	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request	id=0x75dd, seq=3/768, ttl=64 (reply in 164)
164	582.284076	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply	id=0x75dd, seq=3/768, ttl=61 (request in 163)
165	583.284995	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request	id=0x76dd, seq=4/1024, ttl=64 (reply in 166)
166	583.288238	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply	id=0x76dd, seq=4/1024, ttl=61 (request in 165)
167	584.289819	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request	id=0x77dd, seq=5/1280, ttl=64 (reply in 168)
168	584.296140	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply	id=0x77dd, seq=5/1280, ttl=61 (request in 167)
169	585.336635	0c:d5:34:f4:00:00	Private_66:68:00	ARP	60 Who has 10.0.10.10? Tell 10.0.10.1	
170	585.336887	Private_66:68:00	0c:d5:34:f4:00:00	ARP	60 10.0.10.10 is at 00:50:79:66:68:00	
171	587.000288	10.0.10.10	10.0.11.10	TCP	74 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251898 TSecr=0 ...	
172	587.001585	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
173	587.003277	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
174	587.003593	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
175	587.005335	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
176	587.006029	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
177	587.006669	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
178	587.012729	10.0.1.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
179	587.013095	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
180	587.014663	10.0.1.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
181	587.015943	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
182	587.018007	10.0.1.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
183	587.019059	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
184	587.022876	10.0.2.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
185	587.023315	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet	
186	587.024239	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
187	587.026426	10.0.2.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
188	587.028077	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
189	587.032320	10.0.2.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)	
190	587.034633	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
191	587.038276	10.0.11.10	10.0.10.10	TCP	60 29970 → 29969 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0	
192	587.039535	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...	
193	587.044750	10.0.11.10	10.0.10.10	TCP	60 [TCP Port numbers reused] 29970 → 29969 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0	
194	587.045745	10.0.10.10	10.0.11.10	TCP	74 29969 → 29970 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251898 TSecr=0 ...	
195	587.052134	10.0.11.10	10.0.10.10	TCP	60 [TCP Port numbers reused] 29970 → 29969 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0	
196	594.648414	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1	
197	597.028063	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet	
198	607.030666	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet	
199	609.798090	10.0.10.1	224.0.0.9	RIPv2	146 Response	
200	611.659794	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1	
201	617.033846	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet	
202	627.038916	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet	

Рис. 54: Анализ ошибок TCP и ICMP

217	653.274971	10.0.10.10	10.0.11.10	ICMP	98 Echo (ping) request id=0xbccdd, seq=5/1280, ttl=64 (reply in 218)
218	653.280482	10.0.11.10	10.0.10.10	ICMP	98 Echo (ping) reply id=0xbccdd, seq=5/1280, ttl=61 (request in 217)
219	654.315894	0c:d5:34:f4:00:00	Private_66:68:00	ARP	60 Who has 10.0.10.10? Tell 10.0.10.1
220	654.316090	Private_66:68:00	0c:d5:34:f4:00:00	ARP	60 10.0.10.10 is at 00:50:79:66:68:00
221	655.640509	10.0.10.10	10.0.11.10	TCP	74 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251967 TSecr=0 ...
222	655.641074	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
223	655.642653	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
224	655.643832	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
225	655.644947	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
226	655.645358	10.0.10.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
227	655.647136	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
228	655.649759	10.0.4.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
229	655.651017	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
230	655.652586	10.0.4.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
231	655.653749	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
232	655.655059	10.0.4.1	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
233	655.656048	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
234	655.658539	10.0.2.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
235	655.659645	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
236	655.665087	10.0.2.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
237	655.667343	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
238	655.671932	10.0.2.2	10.0.10.10	ICMP	102 Time-to-live exceeded (Time to live exceeded in transit)
239	655.673434	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
240	655.678080	10.0.11.10	10.0.10.10	TCP	60 21281 → 21280 [SYN, ACK] Seq=0 Ack=1 Win=0 Len=0
241	655.678959	10.0.10.10	10.0.11.10	TCP	74 [TCP Retransmission] 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSva...
242	655.685145	10.0.11.10	10.0.10.10	TCP	60 [TCP Port numbers reused] 21281 → 21280 [SYN, ACK] Seq=0 Ack=1 Win=0 Len...
243	655.687478	10.0.10.10	10.0.11.10	TCP	74 21280 → 21281 [SYN] Seq=0 Win=0 Len=0 MSS=1460 TSval=1766251967 TSecr=0 ...
244	655.692145	10.0.11.10	10.0.10.10	TCP	60 [TCP Port numbers reused] 21281 → 21280 [SYN, ACK] Seq=0 Ack=1 Win=0 Len...
245	657.055172	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
246	664.823250	10.0.10.1	224.0.0.9	RIPv2	146 Response
247	667.058254	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
248	669.686874	fe80::ed5:34ff:fef4... ff02::9		RIPng	186 Command Response, Version 1
249	677.062567	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
250	687.063704	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
251	694.832511	10.0.10.1	224.0.0.9	RIPv2	146 Response
252	697.065098	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
253	698.693610	fe80::ed5:34ff:fef4... ff02::9		RIPng	186 Command Response, Version 1
254	707.065061	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
255	717.066048	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
256	718.695616	fe80::ed5:34ff:fef4... ff02::9		RIPng	186 Command Response, Version 1
257	718.836212	10.0.10.1	224.0.0.9	RIPv2	146 Response
258	727.066089	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
259	734.700750	fe80::ed5:34ff:fef4... ff02::9		RIPng	186 Command Response, Version 1
260	737.065635	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
261	747.068017	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
262	754.839342	10.0.10.1	224.0.0.9	RIPv2	146 Response
263	756.708894	fe80::ed5:34ff:fef4... ff02::9		RIPng	186 Command Response, Version 1
264	757.069414	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
265	767.069174	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet

Рис. 55: Настройка OSPFv3 на gw-01

```
msk-nsandryushin-gw-01# configure terminal
msk-nsandryushin-gw-01(config)# router ospf6
msk-nsandryushin-gw-01(config-ospf6)# ospf6 router-id 1.1.1.1
msk-nsandryushin-gw-01(config-ospf6)# exit
msk-nsandryushin-gw-01(config)# interface eth0
msk-nsandryushin-gw-01(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth1
msk-nsandryushin-gw-01(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# interface eth2
msk-nsandryushin-gw-01(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-01(config-if)# exit
msk-nsandryushin-gw-01(config)# exit
msk-nsandryushin-gw-01# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-01# █
```

```
msk-nsandryushin-gw-02# configure terminal
msk-nsandryushin-gw-02(config)# router ospf6
msk-nsandryushin-gw-02(config-ospf6)# ospf6 router-id 2.2.2.2
msk-nsandryushin-gw-02(config-ospf6)# exit
msk-nsandryushin-gw-02(config)# interface eth0
msk-nsandryushin-gw-02(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# interface eth1
msk-nsandryushin-gw-02(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-02(config-if)# exit
msk-nsandryushin-gw-02(config)# exit
msk-nsandryushin-gw-02# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-02#
```

Рис. 57: Настройка OSPFv3 на gw-03

```
msk-nsandryushin-gw-03# configure terminal
msk-nsandryushin-gw-03(config)# router ospf6
msk-nsandryushin-gw-03(config-ospf6)# ospf6 router-id 3.3.3.3
msk-nsandryushin-gw-03(config-ospf6)# exit
msk-nsandryushin-gw-03(config)# interface eth0
msk-nsandryushin-gw-03(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth1
msk-nsandryushin-gw-03(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# interface eth2
msk-nsandryushin-gw-03(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-03(config-if)# exit
msk-nsandryushin-gw-03(config)# exit
msk-nsandryushin-gw-03# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-03#
```

Рис. 58: Настройка OSPFv3 на gw-04

```
msk-nsandryushin-gw-04(config-if)# router ospf6
msk-nsandryushin-gw-04(config-ospf6)# ospf6 router-id 4.4.4.4
msk-nsandryushin-gw-04(config-ospf6)# exit
msk-nsandryushin-gw-04(config)# interface eth0
msk-nsandryushin-gw-04(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# interface eth1
msk-nsandryushin-gw-04(config-if)# ipv6 ospf6 area 0
msk-nsandryushin-gw-04(config-if)# exit
msk-nsandryushin-gw-04(config)# exit
msk-nsandryushin-gw-04# write memory
Note: this version of vtysh never writes vtysh.conf
Building Configuration...
Integrated configuration saved to /etc/frr/frr.conf
[OK]
msk-nsandryushin-gw-04#
```

Рис. 59: Проверка связности OSPFv3

```
VPCS> ping 2001:11::a

2001:11::a icmp6_seq=1 ttl=58 time=28.349 ms
2001:11::a icmp6_seq=2 ttl=58 time=8.277 ms
2001:11::a icmp6_seq=3 ttl=58 time=5.083 ms
2001:11::a icmp6_seq=4 ttl=58 time=9.734 ms
2001:11::a icmp6_seq=5 ttl=58 time=11.764 ms

VPCS> trace 2001:11::a

trace to 2001:11::a, 64 hops max
 1 2001:10::1    4.485 ms   0.880 ms   2.130 ms
 2 2001:1::2     4.217 ms   5.025 ms   3.368 ms
 3 2001:2::2     11.396 ms  3.765 ms   3.199 ms
 4 2001:11::a    5.634 ms   5.830 ms   3.506 ms

VPCS> █
```

Рис. 60: Проверка OSPFv3 на gw-01


```

msk-nsandryushin-gw-01# show ipv6 ospf6 neighbor
Neighbor ID      Pri    DeadTime    State/IfState    Duration I/F[State]
2.2.2.2          1      00:00:38    Full/BDR        00:03:07 eth1[DR]
4.4.4.4          1      00:00:38    Full/BDR        00:01:18 eth2[DR]
msk-nsandryushin-gw-01# show ipv6 ospf6 route
*N IA 2001:1::/64      ::                eth1 00:01:21
*N IA 2001:2::/64      fe80::ed8:daff:fe5f:0 eth1 00:01:21
*N IA 2001:3::/64      fe80::e05:bfff:feca:1 eth2 00:01:21
*N IA 2001:4::/64      ::                eth2 00:01:21
*N IA 2001:10::/64     ::                eth0 00:01:21
*N IA 2001:11::/64     fe80::ed8:daff:fe5f:0 eth1 00:01:21
                        fe80::e05:bfff:feca:1 eth2
msk-nsandryushin-gw-01# █

```

Рис. 61: Анализ ошибок Time Exceeded в Wireshark

358	1112.140491	fe80::ed5:34ff:fef4...	ff02::5	OSPF	90 Hello Packet
359	1113.144336	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=1, hop limit=64 (reply in 362)
360	1113.171209	fe80::ed5:34ff:fef4...	ff02::1:ff00:a	ICMPv6	86 Neighbor Solicitation for 2001:10::a from 0c:d5:34:f4:00:00
361	1113.171474	2001:10::a	fe80::ed5:34ff:fef4...	ICMPv6	86 Neighbor Advertisement 2001:10::a (sol, ovr) is at 00:50:79:66:68:00
362	1113.172124	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x88df, seq=1, hop limit=58 (request in 359)
363	1114.174106	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=2, hop limit=64 (reply in 364)
364	1114.181911	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x88df, seq=2, hop limit=58 (request in 363)
365	1115.185194	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=3, hop limit=64 (reply in 366)
366	1115.190019	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x88df, seq=3, hop limit=58 (request in 365)
367	1116.192889	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=4, hop limit=64 (reply in 368)
368	1116.202029	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x88df, seq=4, hop limit=58 (request in 367)
369	1116.957509	10.0.10.1	224.0.0.9	RIPv2	146 Response
370	1117.150945	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
371	1117.203386	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=5, hop limit=64 (reply in 372)
372	1117.211232	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) reply id=0x88df, seq=5, hop limit=58 (request in 371)
373	1118.760537	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
374	1118.764676	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
375	1118.766478	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
376	1118.767117	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
377	1118.768781	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
378	1118.770136	2001:10::1	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
379	1118.771971	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
380	1118.775962	2001:1::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
381	1118.777337	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
382	1118.781946	2001:1::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
383	1118.783628	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
384	1118.786701	2001:1::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
385	1118.789506	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
386	1118.800382	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
387	1118.800954	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
388	1118.804550	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
389	1118.806643	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
390	1118.809646	2001:2::2	2001:10::a	ICMPv6	174 Time Exceeded (Hop limit exceeded in transit)
391	1118.811238	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
392	1118.816679	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
393	1118.817280	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
394	1118.822895	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
395	1118.824143	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
396	1118.827416	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
397	1122.144008	fe80::ed5:34ff:fef4...	ff02::5	OSPF	90 Hello Packet
398	1127.153389	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
399	1132.147446	fe80::ed5:34ff:fef4...	ff02::5	OSPF	90 Hello Packet
400	1137.157307	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
401	1139.973818	10.0.10.1	224.0.0.9	RIPv2	146 Response
402	1142.150435	fe80::ed5:34ff:fef4...	ff02::5	OSPF	90 Hello Packet
403	1147.160002	10.0.10.1	224.0.0.5	OSPF	78 Hello Packet
404	1147.849347	fe80::ed5:34ff:fef4...	ff02::9	RIPng	186 Command Response, Version 1

Рис. 62: Анализ ошибок Destination Unreachable

253	1129.309378	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=2, hop limit=61 (reply in 255)
254	1120.311087	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=2, hop limit=61 (reply in 254)
255	1120.311508	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=3, hop limit=61 (reply in 257)
256	1121.321192	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=3, hop limit=61 (reply in 256)
257	1121.321635	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=4, hop limit=61 (reply in 259)
258	1122.333098	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=4, hop limit=61 (reply in 258)
259	1122.333452	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=5, hop limit=61 (reply in 261)
260	1122.434537	10.0.11.1	224.0.0.5	OSPF	78 Hello Packet
261	1123.339941	2001:10::a	2001:11::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=5, hop limit=61 (reply in 262)
262	1123.340735	2001:11::a	2001:10::a	ICMPv6	118 Echo (ping) request id=0x88df, seq=5, hop limit=61 (reply in 261)
263	1123.438047	fe80::eb5:37ff:fe3f... ff02::9		RIPng	186 Command Response, Version 1
264	1124.947469	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
265	1124.948248	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
266	1124.954346	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
267	1124.954539	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
268	1124.959105	2001:10::a	2001:11::a	UDP	126 17184 → 17185 Len=64
269	1124.959305	2001:11::a	2001:10::a	ICMPv6	174 Destination Unreachable (Port unreachable)[Malformed Packet]
270	1129.977196	fe80::eb5:37ff:fe3f... ff02::5		OSPF	90 Hello Packet
271	1132.438347	10.0.11.1	224.0.0.5	OSPF	78 Hello Packet
272	1133.385374	10.0.11.1	224.0.0.9	RIPv2	146 Response
273	1139.981233	fe80::eb5:37ff:fe3f... ff02::5		OSPF	90 Hello Packet
274	1142.443023	10.0.11.1	224.0.0.5	OSPF	78 Hello Packet
275	1149.986902	fe80::eb5:37ff:fe3f... ff02::5		OSPF	90 Hello Packet
276	1152.446643	10.0.11.1	224.0.0.5	OSPF	78 Hello Packet

Рис. 63: Продолжение анализа ошибок в Wireshark

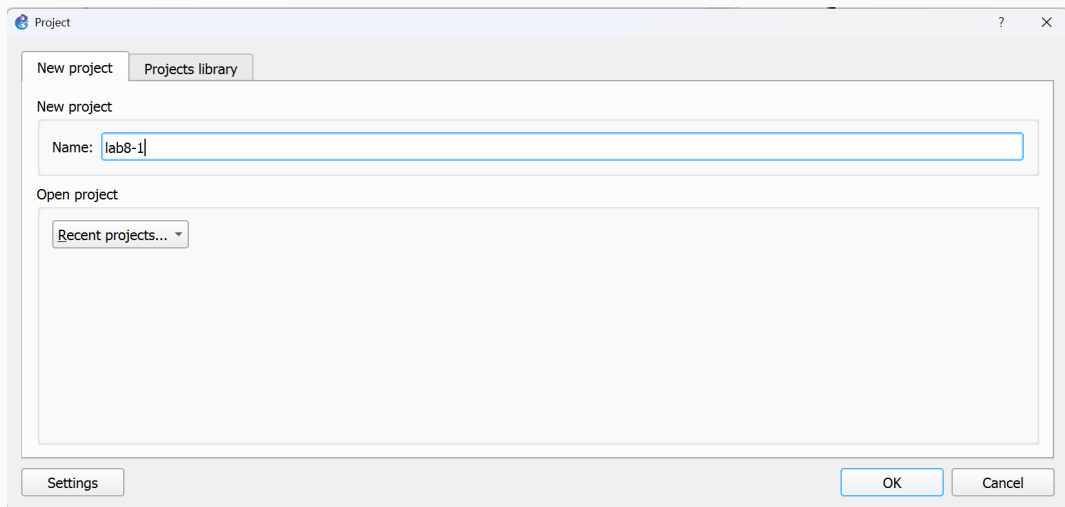


Рис. 64: Создание проекта lab8-1

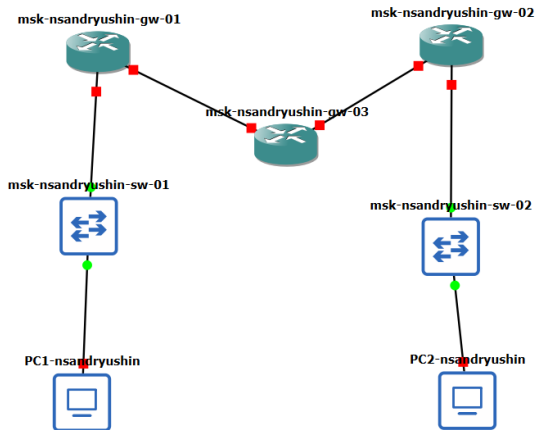


Рис. 65: Топология с VyOS

В результате выполнения лабораторной работы были получены навыки настройки маршрутизации